CONTROLLED ENTERPRISE DOCUMENT

TBT-01.19 | Fall Protection Confined Spaces

MH Construction, INC Enterprise Management System

Document family: TOOLBOX TALK LIBRARY
Document title: TBT-01.19 | Fall Protection Confined Spaces
Version: 1.0 | Lifecycle: Rough Draft

Use only after required functional, technical, field-usability, and approval gates are complete.

Control: Version 1.0 | Status Rough Draft | Author and Chief Editor: Matt Ramsey | Co-Editor: Todd Shoeff | Final Approver: Jeremy Thamert

MH Construction, INC

TOOLBOX TALK: TBT-01.19

TOPIC: Fall Protection Inside Confined Spaces — Retrieval Systems, Vertical Hazards & Pasco Seasonal Factors

TRACK: Track 3 MEP & Interior | Track 1 Civil & Earthwork
OSHA: 29 CFR 1926.1207 / 29 CFR 1926.502 | WISHA: WAC 296-809 / WAC 296-155-24609
CSI Division: 12 / 22 / 31 / 33
Prerequisite: TBT-01.18 Confined Space Entry | TBT-01.17 Fall Protection for Elevated Work
Cross-Reference: MISH-F-32.1 Confined Space Permit | MISH-F-21.1 Fall Protection Inspection

⚠ CRITICAL: A fall inside a confined space is a dual fatality risk. The fall itself may be fatal. The rescue attempt may be fatal. Retrieval system must be rigged BEFORE the entrant descends — not after.

1. DIRECTIVE (Policy Standard)

Every MH Construction, INC entrant descending into a permit-required confined space with a vertical drop of 4 feet or greater SHALL be equipped with a full-body harness connected to a calibrated retrieval system (tripod or davit arm) before entry. This standard applies regardless of entry duration, perceived atmospheric safety, or prior entry history at the same space. WISHA WAC 296-155-24609 (4-foot fall trigger) and OSHA 29 CFR 1926.1207 govern simultaneously. The retrieval system is the primary non-entry rescue mechanism — it must be operable by the attendant alone without entering the space.

2. GENERAL ORDER (Execution Standard)

Fall protection inside a confined space presents unique challenges that do not exist in open-air elevated work. The space geometry restricts harness movement, limits anchor point options, and compresses fall clearance distances. In Pasco's industrial corridor, MH crews encounter four primary confined space geometries requiring fall protection:

Space Type	Fall Protection Method	Seasonal Factor
Manhole / Utility Vault (vertical, narrow)	Tripod with SRL (self-retracting lifeline) — no standard lanyard	Spring: wet ladder rungs from irrigation groundwater. Winter: ice-coated rungs.
Mechanical Pit / Sump (wider, deeper)	Davit arm with retrieval winch — allows lateral positioning	Summer: heat exhaustion accelerates fall risk at depth. Limit rotation to 20 min.
Crawl Space (horizontal, low clearance)	Horizontal lifeline with SRL — traditional vertical retrieval not applicable	All seasons: confined geometry means any incapacitation requires drag-rescue.
PEMB Foundation Pit (open top, deep)	Tripod or structural anchor above opening — WISHA 4-ft trigger applies	Spring/Fall: wind across open pit top creates updraft that destabilizes ladder descent.

The retrieval system must be set up, tested for operation, and confirmed by the attendant BEFORE the entrant clips in and descends. The attendant must be able to raise the entrant from the bottom of the space without assistance. If the retrieval system requires two people to operate, a second attendant must be designated before entry is authorized.

3. KEY HAZARDS & MITIGATION (The "What If")

SCENARIO 1: Manhole Entry — Icy Ladder Rungs in Winter Freezing Fog

Work Type / Season: Div 33 Utilities / Civil | Winter (December–February)

The Hazard: A crew opens a sewer manhole in January. Overnight freezing fog has coated the steel ladder rungs with a thin layer of black ice invisible in the low-light conditions. The entrant descends 8 feet, loses grip on the third rung from the bottom, and falls. The retrieval system was not rigged because the crew assumed the short depth did not require it.

The Mitigation: WISHA 4-foot rule applies to confined space vertical drops — 8 feet is double the trigger. Retrieval tripod must be set up before descent. Inspect every ladder rung by hand before committing weight. If rungs are iced, apply ice melt and wait 5 minutes before re-inspection. Use rubber-soled boots — smooth-soled work boots are a disqualifier on icy rungs.

TBT-01.18 Bridge: TBT-01.18 Scenario 1 identified H2S stratification at the bottom of this same manhole. The entrant who falls is now incapacitated at the bottom of an H2S-contaminated space. The retrieval system is the only way to recover them without a second fatality.

SCENARIO 2: Mechanical Pit — Harness Entanglement in Summer Heat

Work Type / Season: Div 22 Plumbing / Div 23 HVAC | Summer (June–August)

The Hazard: A plumber descends into a 12-foot mechanical pit in 105°F ambient heat. The pit is 6 feet wide with existing pipe runs crossing at multiple elevations. The worker's lanyard catches on a pipe elbow at 6 feet depth, preventing further descent and locking them in a partially suspended position. Heat exhaustion sets in within 8 minutes. The attendant cannot see the entrant due to the pipe obstruction.

The Mitigation: Pre-entry visual inspection of the space must identify all obstruction points. The SRL (self-retracting lifeline) is preferred over a fixed lanyard in pipe-congested spaces because it retracts slack continuously, reducing snagging risk. In summer heat, limit entry rotations to 15 minutes maximum in spaces without forced cooling. The attendant must maintain voice contact every 2 minutes — if no response, initiate retrieval immediately.

TBT-01.18 Bridge: TBT-01.18 Scenario 2 identified O2 depletion in sealed mechanical rooms in summer heat. An entrant who is heat-exhausted in a low-O2 environment loses consciousness faster. The retrieval system must be capable of raising a fully limp 250-lb worker.

SCENARIO 3: Utility Vault — Retrieval System Anchor Point Failure

Work Type / Season: Div 33 Utilities | Fall (September–November)

The Hazard: A crew sets up a retrieval tripod over a utility vault. The tripod legs are placed on asphalt that has been softened by summer heat and not yet re-hardened. During retrieval of an incapacitated entrant, one tripod leg punches through the soft asphalt and the tripod collapses, dropping the entrant back into the vault.

The Mitigation: Tripod leg pads must be used on all soft, asphalt, or unstable surfaces. Before loading the retrieval system, apply body weight to each leg to test surface stability. If the surface is questionable, use a davit arm anchored to a structural element (vehicle hitch, building anchor, concrete barrier) instead of a freestanding tripod.

TBT-01.18 Bridge: TBT-01.18 Scenario 4 identified wildfire CEO concentration in utility vaults in fall. An entrant incapacitated by CEO at the bottom of a vault with a failed retrieval system creates a multi-fatality scenario. Retrieval system integrity is non-negotiable.

SCENARIO 4: PEMB Foundation Pit — Wind Updraft Destabilizes Ladder Descent

Work Type / Season: Div 13 Special Construction / Div 31 Earthwork | Spring/Fall (Wind Season)

The Hazard: A worker begins descending a portable ladder into a 10-foot PEMB foundation pit. A 30 MPH gust crosses the open pit top, creating an updraft that catches the worker's loose jacket and pulls them away from the ladder at 6 feet depth. The retrieval system is rigged but the SRL has not been clipped to the harness D-ring — the worker assumed they would clip in at the bottom.

The Mitigation: The SRL must be clipped to the harness D-ring BEFORE the first step onto the ladder. Not at the bottom. Not halfway down. Before the first step. In wind conditions above 15 MPH, a second worker must spot the ladder at the surface. Loose clothing must be secured or removed before confined space entry.

TBT-01.18 Bridge: TBT-01.18 Scenario 5 identified the PEMB foundation pit as a dual confined space and excavation hazard. TBT-01.16 identified wind gusts in spring and fall as a primary Pasco hazard. This scenario is the direct intersection of both.

SCENARIO 5: Crawl Space — Incapacitation and Horizontal Drag Rescue

Work Type / Season: Div 03 Concrete / Div 06 Wood | All Seasons — Agricultural Sites

The Hazard: A worker enters a crawl space beneath a food processing facility to install plumbing. The space is 18 inches high with a 40-foot horizontal run to the work area. The worker is connected to a horizontal lifeline but the retrieval line is not long enough to reach the work area. The worker is incapacitated by fumigant residue at 35 feet. The attendant cannot drag the worker out because the retrieval line is only 20 feet long.

The Mitigation: Retrieval line length must be measured and confirmed to reach the furthest point of the entrant's work area before entry. For horizontal crawl spaces, a continuous rope with a loop at the entrant's chest harness D-ring allows drag rescue without a mechanical system. The attendant must be physically capable of dragging the entrant's full body weight through the crawl space geometry. If not, a second attendant is required.

TBT-01.18 Bridge: TBT-01.18 Scenario 6 identified fumigant residue in agricultural crawl spaces as a Pasco-specific hazard. Incapacitation from fumigant exposure at 35 feet into a crawl space is an immediate life-safety event — retrieval line length is the difference between recovery and fatality.

SCENARIO 6: Boiler Room Pit — Harness Incompatibility with Confined Space Geometry

Work Type / Season: Div 23 HVAC / Div 22 Plumbing | All Seasons

The Hazard: A worker is assigned to descend into a boiler room pit using a full-body harness. The pit access hatch is 24 inches square. The worker's harness has a dorsal D-ring that catches on the hatch frame during descent, suspending them in the opening. They cannot move up or down. The attendant cannot reach the D-ring to release it.

The Mitigation: Harness selection must account for the access opening dimensions. For hatches narrower than 30 inches, a chest harness with a sternal D-ring may be required instead of a standard dorsal D-ring harness. Pre-entry simulation: the entrant must physically pass through the hatch opening in full PPE and harness before the permit is signed. If they cannot pass through freely, the harness configuration must be changed before entry is authorized.

4. FIELD INSTRUCTION (Immediate Action)

Before any confined space entry with a vertical drop of 4 feet or greater today, execute this 6-point retrieval system check:

- (1) Retrieval system (tripod or davit arm) is set up and tested for operation BEFORE the entrant clips in.
- (2) Retrieval line length is confirmed to reach the furthest point of the entrant's work area.
- (3) Anchor surface is tested for stability — tripod leg pads used on asphalt or soft surfaces.
- (4) Entrant clips SRL to dorsal D-ring BEFORE stepping onto the ladder — not at the bottom.
- (5) Attendant confirms they can raise a fully limp entrant alone. If not, second attendant designated.
- (6) Harness has been physically tested through the access opening in full PPE before permit is signed.

If the retrieval system fails during an active rescue — stop. Do not improvise. Call 911 immediately. Maintain voice contact with the entrant. Do not send a second person into the space without a functioning retrieval system and a confirmed clean atmospheric reading.

5. ATTENDANCE LOG

Foreman: Scan the QR code below to log attendance digitally via MISH-F-10.1 on the MH Dashboard, or use the physical sign-in sheet in the site binder as the fail-safe. File the completed MISH-F-32.1 Confined Space Entry Permit and MISH-F-21.1 Fall Protection Inspection together with this attendance log — all three documents are required for permit-required confined space entry with fall protection.

[INSERT QR CODE HERE — LINK TO MISH-F-10.1 ON MH DASHBOARD]

SIGN-IN SHEET (Hardcopy Fail-Safe)

Project: ________________________ Date: ________________ Foreman: ________________________

Employee Name (Print)	Role (Entrant / Attendant / Supervisor)	Retrieval System Confirmed (Y/N)	Signature